EXPERIMENT 12: Monthly Temperature and Rainfall Visualization

1. Line Plot of Monthly Temperature

Aim

To develop a Python program to create a line plot representing monthly temperature data using Matplotlib.

Algorithm

1. Import Matplotlib.
2. Get month names and temperature values.
3. Create a line plot using plt.plot().
4. Add a title and axis labels.
5. Display the plot.

Program

Writing

```
import matplotlib.pyplot as plt


months = input("Enter month names: ").split()

temperature = list(map(float, input("Enter temperatures: ").split()))


plt.plot(months, temperature, marker='o')

plt.title("Monthly Temperature")

plt.xlabel("Months")

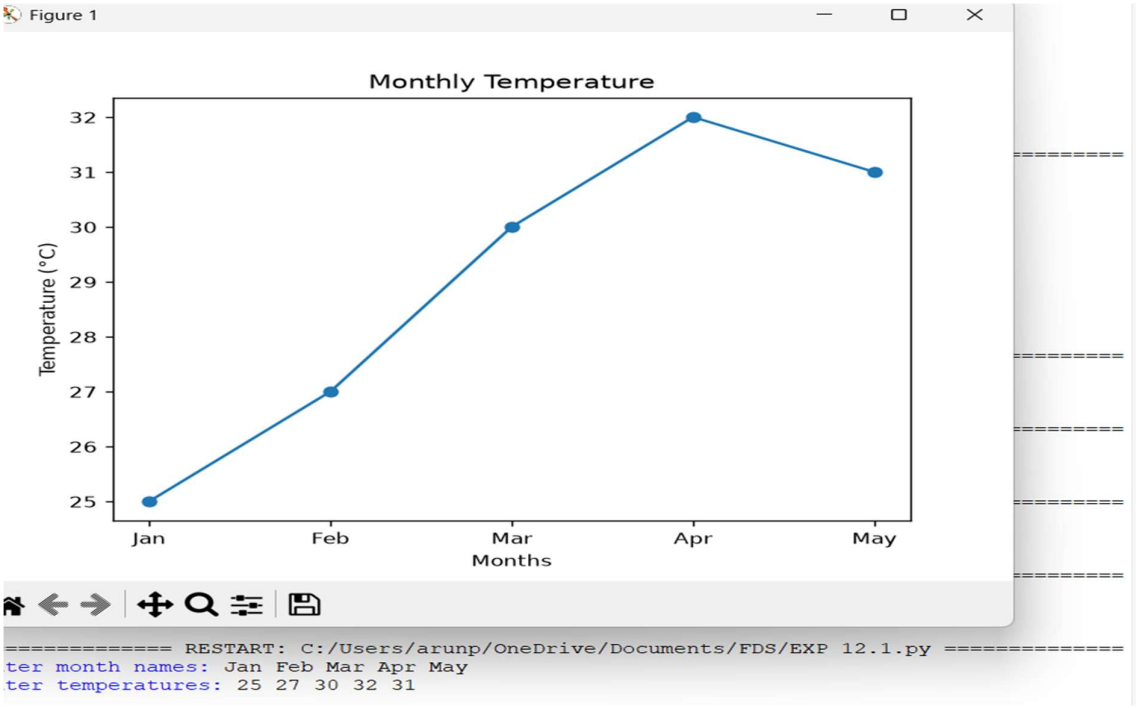
plt.ylabel("Temperature (°C)")

plt.show()
```

Sample Input

Enter month names: Jan Feb Mar Apr May

Enter temperatures: 25 27 30 32 31



Result

The line plot showing monthly temperature variations is generated successfully.

2. Scatter Plot of Monthly Rainfall

Aim

To develop a Python program to create a scatter plot representing monthly rainfall data using Matplotlib.

Algorithm

1. Import Matplotlib.
2. Get month names and rainfall values.
3. Create a scatter plot using plt.scatter().
4. Add a title and axis labels.
5. Display the plot.

Program

Writing

import matplotlib.pyplot as plt

```
months = input("Enter month names: ").split()

rainfall = list(map(float, input("Enter rainfall values in mm: ").split()))


plt.scatter(months, rainfall)

plt.title("Monthly Rainfall")

plt.xlabel("Months")

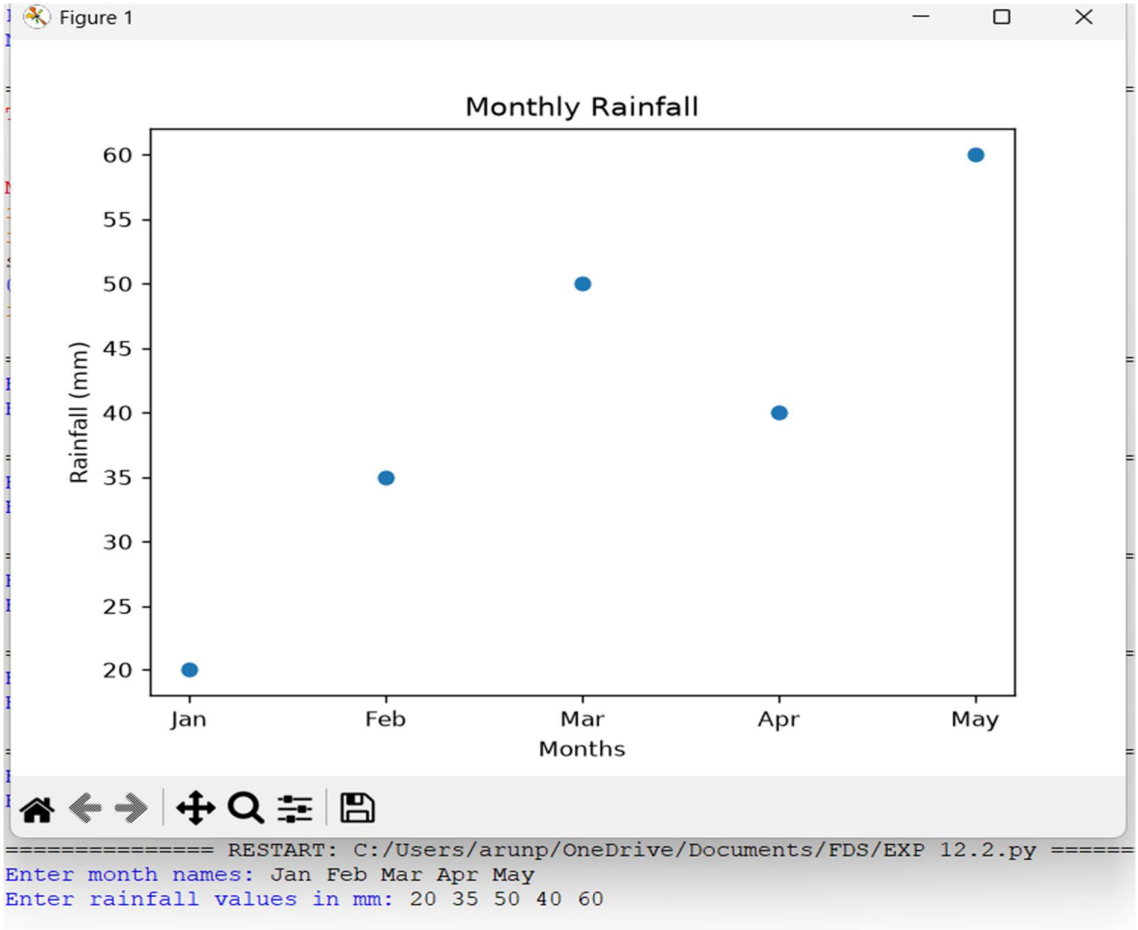
plt.ylabel("Rainfall (mm)")

plt.show()
```

Sample Input

Enter month names: Jan Feb Mar Apr May

Enter rainfall values in mm: 20 35 50 40 60



Result

The scatter plot showing monthly rainfall variations is generated successfully.